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United States Patent	12400867
Kind Code	B2
Date of Patent	August 26, 2025
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Integrated circuit device

Abstract

An integrated circuit device includes a substrate, an isolation feature, a memory cell, and a semiconductor device. The substrate has a cell region, a peripheral region, and a transition region between the cell region and the peripheral region. The isolation feature is in the transition region. A top surface of the isolation feature has a first portion and a second portion lower than the first portion, the second portion of the top surface of the isolation feature is between the cell region and the first portion of the top surface of the isolation feature, and a bottom surface of the isolation feature has a step height directly below the second portion of the top surface of the isolation feature. The is memory cell over the cell region of the substrate. The semiconductor device is over the peripheral region of the substrate.

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Appl. No.: **18/488505**

Filed: **October 17, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240047219 A1	Feb. 08, 2024

Related U.S. Application Data

division parent-doc US 17574414 20220112 US 11854823 child-doc US 18488505
division parent-doc US 16716151 20191216 US 11239089 20220201 child-doc US 17574414

Publication Classification

Int. Cl.: **H01L21/3105** (20060101); **H01L21/308** (20060101); **H01L21/762** (20060101); **H10B41/30** (20230101)

U.S. Cl.:

CPC **H01L21/31056** (20130101); **H01L21/3086** (20130101); **H01L21/76283** (20130101); **H10B41/30** (20230201);

Field of Classification Search

USPC: None

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a divisional application of U.S. patent application Ser. No. 17/574,414, filed Jan. 12, 2022, which is a divisional application of U.S. patent application Ser. No. 16/716,151, filed Dec. 16, 2019, now U.S. Pat. No. 11,239,089, issued Feb. 1, 2022, all of which are herein incorporated by reference in their entireties.

BACKGROUND
(1) The semiconductor integrated circuit (IC) industry has experienced exponential growth over the last few decades. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest

component (or line) that can be created using a fabrication process) has decreased. One advancement implemented as technology nodes shrink, in some IC designs, has been the replacement of the polysilicon gate electrode with a metal gate electrode to improve device performance with the decreased feature sizes.
(2) Super-flash technology has enabled designers to create cost effective and high performance programmable SOC (system on chip) solutions through the use of split-gate flash memory cells. The aggressive scaling of the third generation embedded super-flash memory (ESF3) enables designing flash memories with high memory array density.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIGS. **1**A and **1**B are flow charts of a method for fabricating a semiconductor device in accordance with some embodiments.
(3) FIGS. **2**A to **21**C illustrate the method for fabricating a semiconductor device at different stages in accordance with some embodiments.
(4) FIGS. **22**A to **22**B are cross-sectional views of a semiconductor device in accordance with some embodiments.
(5) FIGS. **23**A to **23**B are cross-sectional views of a semiconductor device in accordance with some embodiments.
(6) FIGS. **24**A to **24**B are cross-sectional views of a semiconductor device in accordance with some embodiments.
DETAILED DESCRIPTION
(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(9) Flash memory can be formed on a bulk silicon substrate and uses various bias conditions to read and write data values. For example, an EFS3 cell—or so-called “third generation SUPERFLASH” cell—includes a pair of symmetric split gate memory cells, each of which includes a pair of source/drain regions with a channel region arranged there between. In the EFS3

architecture, one of the source/drain regions for each of the split gate memory cells is a common source/drain region shared with its neighboring cell, while the other source/drain region is an individual source/drain unique to the cell. Within each split gate cell, a floating gate is arranged over the channel region of the cell, and a control gate is arranged over the floating gate. A select gate is arranged on one side of the floating and control gates (e.g., between an individual source/drain region of the EFS3 cell and a sidewall of the floating and/or control gate). At least one cell is configured to store a variable charge level on its floating gate, wherein the level of this charge corresponds to a data state stored in the cell and is stored in a non-volatile manner so that the stored charge/data persists in the absence of power.

(10) By changing the amount of charge stored on the floating gate, the threshold voltage V.sub.th of the memory cell device can be correspondingly changed. For example, to perform a program operation (e.g., write a logical “0”, program is 0, V.sub.th high) for a cell, the control gate is biased with a high (e.g., at least an order of magnitude higher) voltage relative a voltage applied across the channel region and/or relative to a voltage applied to the select gate. The high bias voltage promotes Fowler-Nordheim tunneling of carriers from the channel region towards the control gate. As the carriers tunnel towards the control gate, the carriers become trapped in the floating gate and alter the V.sub.th of the cell. Conversely, to perform an erase operation (e.g., write a logical “1”, erase is 1, V.sub.th low) for the cell, the erase gate is biased with a high (e.g., at least an order of magnitude higher) voltage relative a voltage applied across the channel region and/or relative to a voltage applied to the control gate. The high bias voltage promotes Fowler-Nordheim tunneling of carriers from the floating gate towards the erase gate, thereby removing carriers from the floating gate and again changing the V.sub.th of the cell in a predictable manner. Subsequently, during a read operation, a voltage is applied to the select gate to induce part of the channel region to conduct. Application of a voltage to the select gate attracts carriers to part of the channel region adjacent to the select gate. While the select gate voltage is applied, a voltage greater than V.sub.th, but less than V.sub.th+ΔV.sub.th, is applied to the control gate (where ΔV.sub.th is a change in V.sub.th due to charge trapped on the floating gate). If the memory cell device turns on (i.e., allows charge to flow), then it is deemed to contain a first data state (e.g., a logical “1” is read). If the memory cell device does not turn on, then it is deemed to contain a second data state (e.g., a logical “0” is read).

(11) Due to the high-voltages involved in performing program and/or erase operations, high energy implants are used in some instances to form the source/drain regions of the flash memory cells. Thus, the source/drain regions of the flash cells can be implanted deeper than that of low-voltage CMOS devices. This additional implant depth can help to reduce current crowding at the substrate surface near edges of the source/drain regions.

(12) Some embodiments of the present disclosure relate to flash memory devices that are formed on a recessed region of a substrate. Although some implementations are illustrated below with regards to split gate flash memory, it will be appreciated that this concept is not limited to split gate flash memory cells, but is also applicable to other types of flash memory cells as well as to other types of semiconductor devices, such as MOSFETs, FinFETs, and the like.

(13) FIGS. **1**A and **1**B are flow charts of a method M for fabricating a semiconductor device at different stages in accordance with some embodiments. FIGS. **2**A to **21**C illustrate the method for fabricating a semiconductor device at different stages in accordance with some embodiments. It is understood that additional steps may be implemented before, during, or after the method M, and some of the steps described may be replaced or eliminated for other embodiments of the method M.

(14) FIG. **2**A is a top view of the semiconductor device according with some embodiments, and FIG. **2**B is a cross-sectional view taking along line B-B of FIG. **2**A. Referring to FIG. **1**A and FIGS. **2**A and **2**B, the method M begins at step S**1** where a substrate **110** is provided, and a pad layer **120** and a hard mask layer **130** are formed on the substrate **110**. In some embodiments, the substrate **110** can be a semiconductor substrate, such as a bulk silicon substrate, a germanium

substrate, a compound semiconductor substrate, or other suitable substrate. The substrate **110** may include an epitaxial layer overlying a bulk semiconductor, a silicon germanium layer overlying a bulk silicon, a silicon layer overlying a bulk silicon germanium, or a semiconductor-on-insulator (SOI) structure. The substrate **110** includes a cell region **112**, a peripheral region **114**, and a transition region **116**. The peripheral region **114** is located at at least one edge of the cell region **112**. For example, the peripheral region **114** surrounds the cell region **112**. The transition region **116** is disposed between the cell region **112** and the peripheral region **114**.

(15) The pad layer **120** may be a thin film comprising silicon oxide formed using, for example, a thermal oxidation process. The pad layer **120** may act as a buffer layer between the substrate **110** and hard mask layer **130**. The pad layer **120** may also act as an etch stop layer for etching the hard mask layer **130** in subsequent process. In some embodiments, the hard mask layer **130** is formed of dielectric material, such as silicon nitride, for example, using low-pressure chemical vapor deposition (LPCVD) or plasma enhanced chemical vapor deposition (PECVD). For example, the pad layer **120** may have a thickness in a range from about 30 angstroms to about 300 angstroms.

(16) Referring to FIG. **1**A and FIG. **3**, where the cross-sectional position of FIG. **3** is the same as the cross-sectional position of FIG. **2**B. The method M proceeds to step S**2** where the substrate **110** is patterned to form plural trenches in the transition and peripheral regions. In the present embodiments, the pad layer **120**, the hard mask layer **130** (referring to FIG. **2**B) are patterned to form a patterned pad layer **120**′ and a patterned hard mask layer **130**′, respectively, and the substrate **110** is patterned to form a trench **116**T in the transition region **116** and at least one trench **114**T in the peripheral region **114**.

(17) For example, a photoresist layer is formed on the hard mask layer **130** (referring to FIG. **2**B) and then patterned by photolithography processes, forming openings in the photoresist layer, such that some regions of the hard mask layer **130** (referring to FIG. **2**B) above the peripheral region **114** and the transition region **116** of the substrate **110** are exposed by the photoresist layer. The exposed portions of the hard mask layer **130** and the underlying pad layer **120** (referring to FIG. **2**B) are etched and removed, and the remaining hard mask layer **130** and the underlying pad layer **120** (referring to FIG. **2**B) are referred to as a patterned hard mask layer **130**′ and a patterned pad layer **120**′. The patterned hard mask layer **130**′ cover the cell region **112** and portions of the peripheral region **114** of the substrate **110** and exposes portions of the peripheral region **114** and the transition region **116** of the substrate **110**. The exposed portions of the peripheral region **114** and the transition region **116** of the substrate **110** are then etched using the patterned hard mask layer **130**′ as etch mask, for example, by dry etching such as reactive-ion etching (RIE) or by wet etching using a liquid substrate etchant. For example, gas etchants like HBr, and Cl.sub.2, may be used in the etching the substrate **110**, and the hard mask layer **130**′ may have a higher etch resistance to the etchant than that of the substrate **110**. Through the etching process, trenches **114**T and **116**T are formed.

(18) Referring to FIG. **1**A and FIGS. **4**A and **4**B, where FIG. **4**A is a top view of the semiconductor device according with some embodiments, and FIG. **4**B is a cross-sectional view taking along line B-B of FIG. **4**A. The method M proceeds to step S**3** where isolation features are formed in the trenches in the transition and peripheral regions. In the present embodiments, isolation features **144** and **146** are formed in the trenches **114**T and **116**T, respectively. The isolation features **144** and **146** define an active region **114***a* in the peripheral region **114**. It is noted that the number of the isolation feature **144** can be plural in some other embodiments, and the plural isolation features **144** may define plural active regions **114***a* in the peripheral region **114**. In some embodiments, the isolation features **144** and **146** are made of silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), or other low-K dielectric materials. In the present embodiments, the opening sizes of the trenches **114**T and **116**T may result in loading effect in etching process, such that a bottom of the trench **116**T is lower than a bottom of the trench **114**T, and therefore a bottom surface **146**B of the isolation feature **146** is lower than a bottom surface **144**B of the isolation feature **144**.

(19) In some embodiments, a dielectric material may be formed over the structure of FIG. **3** by suitable process, such as a high-density-plasma (HDP) chemical vapor deposition (CVD) process, a sub-atmospheric CVD (SACVD) process, a high aspect-ratio process (HARP), a spin-on-dielectric (SOD) process or other suitable process. The dielectric material may overfill the trenches **114**T and **116**T. In some embodiments, a liner oxide (not shown) may be formed optionally in advance. In some embodiments, the liner oxide may be a thermal oxide. A chemical mechanical polish (CMP) process is then performed to remove the excess dielectric material outside the trenches **114**T and **116**T, and the CMP process may level the top surface of the dielectric material to the top surfaces of the patterned hard mask layer **130**′, thereby forming the isolation features **144** and **146**.
(20) Referring to FIG. **1**A and FIG. **5**, where the cross-sectional position of FIG. **5** is the same as the cross-sectional position of FIG. **4**B. The method M proceeds to step S**4** where a pad layer, a hard mask layer, and a pad layer are formed over the substrate. In the present embodiments, a pad layer **150**, a hard mask layer **160**, and a pad layer **170** are formed over the substrate **110** in a sequence. The pad layers **150** and **170** may be formed of dielectric material, such as an oxide layer. The pad layer **150** may act as a buffer layer. The hard mask layer **160** is formed over the pad layer **150**. In some embodiments, the hard mask layer **160** is formed of dielectric material, such as silicon nitride (SiN) or other suitable materials. The pad layer **170** is formed over the hard mask layer **160**. The pad layer **150**, the hard mask layer **160**, and the pad layer **170** are formed to be protection layers for the peripheral region **114** in the following etching process.
(21) Referring to FIG. **1**A and FIGS. **6**A and **6**B, where FIG. **6**A is a top view of the semiconductor device, FIG. **6**B is a cross-sectional view taking along line B-B of FIG. **6**A. The method M proceeds to step S**5** where the pad layers and the hard mask layers are patterned to expose a cell region of the substrate **110**. In the present embodiments, the layers **150** to **170**, the hard mask layer **130**′, and the pad layer **120**′ are patterned by suitable etching processes, such that portions of the pad layer **170**, the hard mask layer **160**, the pad layer **150**, the hard mask layer **130**′, and the pad layer **120**′ over the cell region **112** are removed. For example, a photoresist layer is formed on the pad layer **170** (referring to FIG. **5**) and then patterned by photolithography processes, forming openings in the photoresist layer, such that some regions of the pad layer **170** (referring to FIG. **5**) above the cell region **112** of the substrate **110** is exposed by the photoresist layer. The patterning process includes etching the exposed portions of the pad layer **170** and the underlying layers **150** and **160**, **130**′ and **120**′ (referring to FIG. **5**). After the etching processes, the cell region **112** of the substrate **110** is exposed. The etching processes may also remove the pad layer **170** (referring to FIG. **5**) over the peripheral region **114** and the transition region **116**. The remaining hard mask layer **160** covers the peripheral region **114** and a portion of the transition region **116**. In some embodiments, a portion of the isolation feature **146** not covered by the hard mask layer **160** may be etched. For example, the etching process may smooth the corner of the isolation feature **146** not covered by the hard mask layer **160**.
(22) Referring to FIG. **1**A and FIG. **7**, where the cross-sectional position of FIG. **7** is the same as the cross-sectional position of FIG. **6**B. The method M proceeds to step S**6** where the cell region of the substrate is recessed. In the present embodiments, the cell region **112** of the substrate **110** is recessed. For example, a surface layer of the exposed region of the substrate **110** not covered by the hard mask layer **160** is oxidized using, for example, wet oxidation. Thereafter, the oxidized surface layer is removed from the substrate **110** using, for example, wet etching, dry etching, or a combination of wet etching and dry etching. Etchants in the wet and/or dry etching process may include HF or other suitable etchants. The removal of oxidized surface layer results in the recess **112**R in the cell region **112**. For example, a top surface **112**S of the cell region **112** is lower than a top surface **114**S of the peripheral region **114**. In some embodiments, the depth of the recess **112**R is about 50 Angstroms to about 2000 Angstroms. In some embodiments, the hard mask layer **160** has a higher etch resistance to an etchant used in the etching process than that of the oxidized surface layer, thereby protecting underlying layers in the peripheral region **114** from being etched.

In some embodiments, the portion of the isolation feature **146** not covered by the hard mask layer **160** may be further etched in this process. For example, a top surface **146**TB of the portion of the isolation feature **146** not covered by the hard mask layer **160** may be higher than the top surface **112**S of the cell region **112** but lower than a top surface **146**TA of a portion of the isolation feature **146** covered by the hard mask layer **160**. That is, the top surface **146**T of the isolation feature **146** has a step.
(23) Referring to FIG. **1**A and FIG. **8**, where the cross-sectional position of FIG. **8** is the same as the cross-sectional position of FIG. **7**. The method M proceeds to step S**7** where a tunneling film is formed over the cell region of the substrate. In the present embodiments, a tunneling film **180** is then conformally formed over the substrate **110**. In some embodiments, the tunneling film **180** may include, for example, a dielectric material such as silicon dioxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), oxynitrides (SiON), high-k materials, other non-conductive materials, or combinations thereof. The tunneling film **180** may be formed using thermal oxidation, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), ozone oxidation, other suitable processes, or combinations thereof. The thermal oxidation may result in the tunneling film **180** with a uniform thickness over the cell region **112**. In some embodiments, the formation of the tunneling film **180** may also form a dielectric layer **188** over the hard mask layer **160**. In some embodiments, the dielectric layer **188** and the tunneling film **180** have the same material.
(24) Referring to FIG. **1**A and FIG. **9**, where the cross-sectional position of FIG. **9** is the same as the cross-sectional position of FIG. **8**. The method M proceeds to step S**8** where a floating gate film, a pad layer, and a hard mask layer are formed over the substrate. In the present embodiments, a floating gate film **190** is conformally formed over the structure in FIG. **8** (i.e., over the tunneling film **180**, the isolation feature **146**, and the dielectric layer **188**). The floating gate film **190** may include polysilicon formed through, for example, low pressure CVD (LPCVD) methods, CVD methods and PVD sputtering methods employing suitable silicon source materials. The floating gate film may be deposited with desired thickness for floating gates. For example, a thickness of the floating gate film **190** is in a range of about 50 angstroms to about 150 angstroms. If the thickness of the floating gate film **190** is greater than about 150 angstroms, thick floating gates would be formed, and control gates later formed over the floating gates would have a higher top surface, which makes it difficult to integrate the fabrication process of the memory devices into a high-k metal gate (HKMG) process of logic devices. If the thickness of the floating gate film **190** is less than about 50 angstroms, the formed memory devices may have poor storage ability. If desired, the floating gate film **190** may be ion implanted to the desired conductive type. For example, the floating gate film **190** may be in-situ doped. The floating gate film **190** may include other gate electrode material such as metal, metal alloys, single crystalline silicon, or combinations thereof.
(25) After the formation of the floating gate film **190**, another pad layer **200** is conformally formed over the floating gate film **190**, and another hard mask layer **210** is conformally formed over the pad layer **200**. The pad layer **200** may be formed of dielectric material, such as an oxide layer. The pad layer **200** may serve as a buffer layer between the floating gate film **190** and the hard mask layer **210**. The hard mask layer **210** can be formed of dielectric material, such as silicon nitride (SiN) or other suitable materials.
(26) Referring to FIG. **1**A and FIGS. **10**A to **10**C, where FIG. **10**A is a top view of the semiconductor device, FIG. **10**B is a cross-sectional view taking along line B-B of FIG. **10**A, and FIG. **10**C is a cross-sectional view taking along line C-C of FIG. **10**A. The method M proceeds to step S**9** where the substrate is patterned to form plural trenches in the cell region. In the present embodiments, the hard mask layer **210**, the pad layer **200**, the floating gate film **190**, the tunneling film **180**, and the substrate **110** of FIG. **9** are patterned, so as to form trenches **112**T in the cell region **112** and a trench **116**T′ in the transition region **116**. In some embodiments, a width of the trench **116**T′ taking along line B-B of FIG. **10**A may be greater than a width of the trench **116**T′

taking along line C-C of FIG. **10**A. In the present embodiments, the trenches **116**T′ exposes an upper part of a sidewall of the isolation features **146** and a part of the surface **146**TB of the isolation features **146** uncovered by the hard mask layer **160**. A lower part of the sidewall of the isolation features **146** may be covered by a protrusion portion **116***p* of the substrate **110**. In some embodiments, the protrusion portion **116***p* has a tapered shape. For example, the protrusion portion **116***p* tapered upward. A peak of the protrusion portion **116***p* is lower than the top surface **146**TB and **146**TA of the isolation feature **146**. In some embodiments, the peak of the protrusion portion **116***p* is substantially leveled with the top surface **112**S of the cell region **112** of the substrate **110**.
(27) For example, a photoresist layer is formed on the hard mask layer **210** (referring to FIG. **9**) and then patterned by photolithography processes, forming openings in the photoresist layer, such that regions of the hard mask layer **210** (referring to FIG. **9**) are exposed by the photoresist layer. The hard mask layer **210** is patterned by etching the exposed portions of the hard mask layer **210** (referring to FIG. **9**). After the patterning process, the patterned hard mask layer **210**′ covers portions of the pad layer **200** (referring to FIG. **9**) and exposes portions of the pad layer **200** (referring to FIG. **9**).
(28) The exposed portions of the pad layer **200**, the floating gate film **190**, the underlying tunneling layer **180**, and the substrate **110** are then etched using the patterned hard mask layer **210**′ as etch mask by plural dry etching processes, such as reactive-ion etching (RIE). The dry etching processes may use various gas etchants. For example, gas etchants like HBr, Cl.sub.2, CF.sub.4, and/or CHF.sub.3 may be used during the dry etching processes. The patterned hard mask layer **210**′ may have a higher etch resistance to the etchants than that of the pad layer **200**, the floating gate film **190**, the tunneling film **180**, and the substrate **110** (referring to FIG. **9**), thereby protecting underlying layers in the cell region **112** from being etched. The dielectric layer **188** may be removed by the dry etching processes. The hard mask layer **160** may have a higher etch resistance to the etchants than that of the dielectric layer **188**, the floating gate film **190**, the tunneling film **180**, and the substrate **110** (referring to FIG. **9**), thereby protecting underlying layers in the peripheral region **114** from being etched. Through the etching processes, trenches **112**T and **116**T′ are formed.
(29) In the present embodiments, the hard mask layer **210** of FIG. **9** is patterned to be a patterned hard mask layer **210**′, the pad layer **200** of FIG. **9** is patterned to be a patterned pad layer **200**′, the floating gate film **190** of FIG. **9** is patterned to be a patterned floating gate film **190**′, the tunneling film **180** of FIG. **9** is patterned to be a patterned tunneling film **180**′, and the substrate **110** is patterned to include plural base portions **112***b* in the cell region **112** of the substrate **110**. The base portions **112***b* are separated from each other by the trench(es) **112**T. The tunneling film **180**′ is disposed over the substrate **110**, the floating gate film **190**′ is disposed over the tunneling film **180**′, the patterned pad layer **200**′ is disposed over the floating gate film **190**′, and the hard mask layer **210**′ is disposed over the patterned pad layer **200**′. Through the etching processes, portions of the hard mask layer **210**′, the pad layer **200**′, the floating gate film **190**′, and the tunneling film **180**′ of FIG. **9** over the peripheral region **114** are removed.
(30) Referring to FIG. **1**B and FIGS. **11**A to **11**C, where FIG. **11**A is a top view of the semiconductor device, FIG. **11**B is a cross-sectional view taking along line B-B of FIG. **11**A, and FIG. **11**C is a cross-sectional view taking along line C-C of FIG. **11**A. The method M proceeds to step S**10** where isolation features are formed in the trenches in the cell region. In the present embodiments, isolation features **222** and **226** are formed in the trenches **112**T and **116**T′, respectively. The isolation features **222** defines plural active regions **112***a* (e.g., portions of the base portions **112***b*) in the cell region **112**. The isolation feature **226** isolates active regions **112***a* in the cell region **112** from the active regions **114***a* in the peripheral region **114**. According to the profile of the trench **116**T′, a width of the isolation feature **226** taking along line B-B of FIG. **11**A may be greater than a width of the isolation feature **226** taking along line C-C of FIG. **11**A. In the present embodiments, the substrate **110** includes a protruding portion **116***p* between the isolation features

146 and **226** in the transition region **116**. The protruding portion **116***p* tapers upward. In some embodiments, the isolation features **222** and **226** are made of silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), or other low-K dielectric materials.

(31) In some embodiments, a dielectric material may be formed over the structure of FIGS. **10**B and **10**C by suitable process, such as a high-density-plasma (HDP) chemical vapor deposition (CVD) process, a sub-atmospheric CVD (SACVD) process, a high aspect-ratio process (HARP), a spin-on-dielectric (SOD) process or other suitable process. The dielectric material may overfill the trenches **112**T and **116**T′. In some embodiments, a liner oxide (not shown) may be formed optionally in advance. In some embodiments, the liner oxide may be a thermal oxide. A chemical mechanical polish (CMP) process is then performed to remove the excess dielectric material outside the trenches **112**T and **116**T′, and the CMP process may level the top surface of the dielectric material to the top surfaces of the patterned hard mask layer **160**, thereby forming the isolation features **222** and **226**. In some embodiments, the dielectric layer **188** (e.g., oxide layer) over the hard mask layer **160** is removed by the CMP process.

(32) The isolation features **222** may be in contact with the base portions **112***b* of the substrate **110**, the patterned tunneling film **180′**, the patterned floating gate film **190′**, the patterned pad layer **200′**, and the patterned mask layer**210′**. In the present embodiments, the isolation feature **226** is in contact with an upper part of the sidewall of the isolation features **146** and a part of the surface of the isolation features **146** uncovered by the hard mask layer **160**.

(33) In the present embodiments, a bottom of the trench **116**T′ is higher than a bottom of the trench **116**T, such that a bottom surface **226**B of the isolation feature **226** is higher than a bottom surface **146**B of the isolation feature **146**. In some other embodiments, the bottom surface **226**B of the isolation feature **226** may not be higher than the bottom surface **146**B of the isolation feature **146** in some embodiments. For example, the bottom of the trench **116**T′ may be lower than the bottom of the trench **116**T, such that the bottom surface **226**B of the isolation feature **226** is lower than the bottom surface **146**B of the isolation feature **146**. Alternatively, in some other embodiments, the bottom of the trench **116**T′ may be substantially level with the bottom of the trench **116**T, such that the bottom surface **226**B of the isolation feature **226** is substantially level with the bottom surface **146**B of the isolation feature **146**.

(34) Referring to FIG. **1**B and FIGS. **12**A to **12**B, where the cross-sectional positions of FIGS. **12**A to **12**B are the same as the cross-sectional position of FIGS. **11**B to **11**C. The method M proceeds to step S**10** where isolation features in the cell region are recessed. In the present embodiments, the isolation features **222** and **226** are recessed by a wet etch process. For example, liquid etchants, such as HF, are dispensed onto the structure of FIGS. **11**A-**11**C, thereby etching the isolation features **222** and **226**. The patterned hard mask layer **210′** and the patterned hard mask layer **160** have a higher etch resistance to the etchant than that of the isolation features **222** and **226**, such that the isolation features **144** and **146** under the hard mask layers **160** are prevented from being etched, and the layers **180′** to **200′** under the masks **210′** are prevented from being etched. In some other embodiments, a portion of the isolation feature **146** uncovered by the hard mask layers **160** may be etched by the wet etch process.

(35) Referring to FIG. **1**B and FIGS. **13**A to **13**B, where the cross-sectional positions of FIGS. **13**A to **13**B are the same as the cross-sectional position of FIGS. **12**A to **12**B. The method M proceeds to step S**12** where the hard mask layers are removed. In the present embodiments, the patterned hard mask layer **210′** and the patterned hard mask layer **160** are removed, and the pad layers **200′** and **150** are exposed. The removal method may include suitable etching back process, for example, using phosphoric acid as etchant. The pad layers **200′** and **150** may have a higher etch resistance to the etching process than that of the patterned mask layers **210′** and **160**, such that the pad layers **200′** and **150** may protect the underlying layers from being etched in the etching process. In some embodiments, the isolation features **222** and **226** may have a higher etch resistance to the etching process than that of the patterned mask layers **210′** and **160**, such that the isolation features **222** and

226 remains intact after the etching process.
(36) Referring to FIG. **1**B and FIGS. **14**A to **14**B, where the cross-sectional positions of FIGS. **14**A to **14**B are the same as the cross-sectional position of FIGS. **13**A to **13**B. The method M proceeds to step S**13** where the isolation features are recessed. In the present embodiments, the isolation features **222** and **226** are recessed by a wet etch process. For example, liquid etchants, such as HF, are dispensed onto the structure of FIGS. **13**A and **13**B, thereby etching the isolation features **222** and **226**. The pad layers **200′** and **150** (referring to FIGS. **13**A to **13**B) may be removed by the wet etching process. The hard mask layer **130′** and the floating gate film **190′** may have a higher etch resistance to the etchant than that of the isolation features **222** and **226** and the pad layers **200′** and **150** (referring to FIGS. **13**A to **13**B), such that the tunneling film **180′** under the a floating gate film **190′** is prevented from being etched, and the pad layer **120′** under the hard mask layer **130′** is prevented from being etched. In some embodiments, a portion of the isolation feature **226** adjacent to the isolation feature **146** is etched by the wet etch process. After the recessing, the floating gate film **190′** protrudes from a top surface of the isolation features **222** and **226**. Recessing the isolation features **222** may enhance a coupling ratio between floating gates and control gates that are subsequently formed.
(37) Referring to FIG. **1**B and FIGS. **15**A and **15**B, where the cross-sectional positions of FIGS. **15**A and **15**B are respectively the same as the cross-sectional position of FIGS. **14**A to **14**B. The method M proceeds to step S**14** where a dielectric film, a control gate film, and a hard mask layer are formed over the substrate. In the present embodiments, a dielectric film **310** is conformally formed over the structure of FIGS. **14**A to **14**B. In some embodiments, the dielectric film **310** and the tunneling film **180′** may have the same or different materials. That is, the dielectric film **310** may include, for example, a dielectric material such as silicon dioxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), oxynitrides (SiON), high-k materials, other non-conductive materials, or combinations thereof. The dielectric film **310** may be formed using chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), ozone oxidation, other suitable processes, or combinations thereof.
(38) A control gate film **320** is conformally formed over the dielectric film **310**. The control gate film **320** may include polysilicon formed through, for example low pressure CVD (LPCVD) methods, CVD methods and PVD sputtering methods employing suitable silicon source materials. If desired, the control gate film **320** may be ion implanted to the desired conductive type. It is to be appreciated that the control gate film **320** may include other gate electrode material such as metal, metal alloys, single crystalline silicon, or combinations thereof.
(39) A hard mask layer **330** is conformally formed over the control gate film **320**. The hard mask layer **330** may include single layer or multiple layers. In some embodiments, the hard mask layer **330** includes SiN/SiO.sub.2/SiN stacked layers or other suitable materials. In some embodiments, the hard mask layer **330** may be formed using chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), ozone oxidation, other suitable processes, or combinations thereof.
(40) Referring to FIG. **1**B and FIGS. **16**A to **16**D, where FIG. **16**A is a top view of the semiconductor device, FIG. **16**B is a cross-sectional view taking along line B-B of FIG. **16**A, FIG. **16**C is a cross-sectional view taking along line C-C of FIG. **16**A, and FIG. **16**D is a cross-sectional view taking along line D-D of FIG. **16**A. The method M proceeds to step S**15** where the dielectric film, the control gate film, and the hard mask layer are patterned to form gate stacks over the cell region of the substrate. In the present embodiments, the hard mask layer **330**, the control gate film **320**, the dielectric film **310**, the floating gate film **190′**, and the tunneling film **180′** of FIGS. **15**A and **15**B are patterned to form plural gate stacks **300** over the cell region **112** of the substrate **110** and a semiconductor stack **300′** over the peripheral region **114** and the transition region **116**. The hard mask layer **330** of FIGS. **15**A and **15**B can be patterned to form hard masks **332** and a hard mask layer **334**. The control gate film **320** of FIGS. **15**A and **15**B can be patterned to form control

gates **322** and a control gate layer **324**. The dielectric film **310** of FIGS. **15**A and **15**B can be patterned to form dielectric layers **312** and a dielectric layer **314**. The floating gate film **190**′ of FIGS. **15**A and **15**B can be patterned to form floating gates **192**. In some embodiments, the floating gate **192** may be thinner than the control gate **322**. The tunneling film **180**′ of FIGS. **15**A and **15**B can be patterned to form tunneling layers **182**.

(41) In some embodiments, at least one of the gate stacks **300** includes a tunneling layer **182**, a floating gate **192**, a dielectric layer **312**, a control gate **322**, and a hard mask **332**. At least one of the gate stacks **300** may further include a pair of spacers **340** disposed over the floating gate **192** and on opposite sides of the dielectric layer **312**, the control gate **322**, and the hard mask **332**. For clarity, the spacers **340** are illustrated in FIGS. **16**B and **16**C and are omitted in FIG. **16**A. In some embodiments, the spacer **340** includes an inner silicon oxide layer, a middle silicon nitride layer, and an outer silicon oxide layer. The semiconductor stack **300**′ may include the dielectric layer **314**, the control gate layer **324** over the dielectric layer **314**, and the hard mask layer **334** over the control gate layer **324**.

(42) Referring to FIG. **1**B and FIGS. **17**A and **17**B, where the cross-sectional positions of FIGS. **17**A and **17**B are respectively the same as the cross-sectional position of FIGS. **16**B and **16**C. The method M proceeds to step S**16** where spacers are formed on opposite sides of the gate stacks. In the present embodiments, spacers **345** are formed on opposite sides of the gate stacks **300**. In some embodiments, the spacers **345** are high temperature oxide layer or other suitable dielectric layers. In some embodiments, a dielectric film may be conformally formed over the structure of FIGS. **16**A to **16**C, and an etching process (e.g., dry etch process) is performed to remove the horizontal portions of the dielectric film to form the spacers **345**.

(43) Referring to FIG. **1**B and FIGS. **18**A and **18**B, where the cross-sectional positions of FIGS. **18**A and **18**B are respectively the same as the cross-sectional position of FIGS. **17**A and **17**B. The method M proceeds to step S**17** where source regions are formed between two adjacent gate stacks. In the present embodiments, the spacers **345** between adjacent two gate stacks **300** are removed, and source regions SR are formed between two adjacent gate stacks **300**. For example, a patterned photoresist layer is formed by a combination of spin coating, exposing and developing processes to expose areas of the substrate **110** between adjacent gate stacks **300**. The exposed spacers **345** are then removed, and ions are implanted into the areas to form the source regions SR. A common source (CS) dielectric layer SRD is formed over the source region SR. The CS dielectric layer SRD may be a dielectric isolation structure and may be formed by oxidizing the substrate **110**, other suitable processes, or combinations thereof. The patterned photoresist layer is then removed, and the removal method may be performed by solvent stripping or plasma ashing, for example.

(44) Referring to FIG. **1**B and FIGS. **19**A to **19**C, where FIG. **19**A is a top view of the semiconductor device, FIG. **19**B is a cross-sectional view taking along line B-B of FIG. **19**A, and FIG. **19**C is a cross-sectional view taking along line C-C of FIG. **19**A. The method M proceeds to step S**18** where erase gates and select gates are formed on opposite sides of the gate stacks. In the present embodiments, plural select gate dielectric layers **352** and plural select gates (or word lines) **362** are formed on first sides of the gate stacks **300**, and plural erase gates **364** are formed on second sides of the gate stacks **300**. For example, a dielectric layer is formed over the substrate **110**, for example, by a thermal oxidation process, chemical vapor deposition, or atomic layer deposition, a conductive layer is deposited over the dielectric layer, and then the conductive layer is patterned or etched back. Then, plural hard masks **370** are respectively formed over the patterned conductive layer, and another etching process is formed to pattern the patterned conductive layer and the dielectric layer using the hard masks **370** as masks to form the erase gates **364**, the select gates **362**, and the select gate dielectric layers **352**. In some embodiments, the erase gates **364** and the select gates **362** may be made of polysilicon or other suitable materials. If desired, the erase gates **364** and the select gates **362** may be ion implanted to the desired conductive type. For example, the erase gates **364** and the select gates **362** may be in-situ doped. In some embodiments, the select gate

dielectric layers **352** may include silicon oxide, silicon nitride, silicon oxynitride, other non-conductive materials, or the combinations thereof.

(45) Referring to FIG. **1**B and FIGS. **20**A and **20**B, where the cross-sectional positions of FIGS. **20**A and **20**B are respectively the same as the cross-sectional position of FIGS. **19**B and **19**C. The method M proceeds to step S**19** where semiconductor devices are formed in the peripheral and transition regions. In the present embodiments, the semiconductor stack **300′** of FIGS. **19**B and **19**C are removed to expose the patterned mask layers **130′** (see FIGS. **19**B and **19**C) and the isolation feature **144**. A portion of the semiconductor stack **300′** (which is referred to as the semiconductor stack **300″** hereinafter) remains over the isolation feature **226** after the removing process. The patterned pad layer **120′** and the patterned hard mask layer **130′** (see FIGS. **19**A and **19**B) are then removed to expose the substrate **110** in the peripheral region **114**. Then, the isolation features **146** and **144** are recessed until the top surfaces of the isolation features **146** and **184** are substantially flush with the substrate **110** in the peripheral region **114**. At least one semiconductor device **400** is formed over the substrate **110** in the peripheral region **114** and at least one dummy semiconductor device **400′** is formed over the isolation feature **146**. In some embodiments, the semiconductor device **400** can be a transistor (such as a high-K metal gate (HKMG) transistor, and/or a logic transistor), and the present disclosure is not limited in this respect. In some embodiments, the dummy semiconductor device **400′** and the semiconductor device **400** are made of the same materials.

(46) In some embodiments, one or more ion implantation processes are performed to the substrate **110**, thereby forming drain regions DR in the cell region **112** and source/drain regions **400**SD in the peripheral region **114**. The drain regions DR and the source/drain regions **400**SD may be formed be the same or different ion implantation processes. In some embodiments, the gate stack **300** and the select gate **362** are disposed in a position between the source region SR and the drain region DR, and drain regions DR are respectively disposed adjacent to the select gates **362**.

(47) Referring to FIG. **1**B and FIGS. **21**A to **21**C, where FIG. **21**A is a top view of the semiconductor device, FIG. **21**B is a cross-sectional view taking along line B-B of FIG. **21**A, and FIG. **21**C is a cross-sectional view taking along line C-C of FIG. **21**A. The method M proceeds to step S**20** where an etch stop layer and an interlayer dielectric layer are formed. In the present embodiments, an etching stop layer **510** is conformally formed over the structure of FIGS. **20**A and **20**B, and an interlayer dielectric (ILD) **520** is formed over the etching stop layer **510**. Then, a chemical mechanical polish (CMP) process is performed to level the top surface of the ILD **520** with the top surfaces of the erase gates **364**, the control gates **322**, the select gates **362** of the memory cells **10** and a top surface of a gate stack **410** of the semiconductor device **400**. In some embodiments, the gate stack **410** may include a gate dielectric, a work function metal layer over the gate dielectric, a metal over the work function metal layer. In some embodiments, the gate stack **410** may include a metal. As such, plural memory cells **10** are formed. At least one of the memory cells **10** includes two gate stacks **300**, one erase gate **364**, two select gate **362**, one source region SR, and two drain region DR. Two adjacent memory cells **10** share one drain region DR.

(48) In FIGS. **21**A to **21**C, the floating gates **192** of the memory cells **10** are formed without being planarized, such that the floating gates **192** are prevented from dishing and erosion issue caused by the planarization process, which in turn will prevent floating gates **192** in array center from being over-polished, and thereby improving the thickness uniformity of the floating gates in the array center and array edge. Therefore, the floating gates **192** of the memory cells **10** in the center and edges of the cell regions **112** have substantially the same thickness. In some embodiments, the tunneling layers **182** are formed by oxidation and therefore have a uniform thickness. That is, the tunneling layers **182** have substantially the same thickness. Through the configuration, the memory cells **10** may have substantially the same electrical characteristics, which results in high yield rate. The term “substantially” as used herein may be applied to modify any quantitative representation which could permissibly vary without resulting in a change in the basic function to which it is

related. It is noted that the number of the memory cells **10** in FIGS. **21**A to **21**C is illustrative, and should not limit the present disclosure. In some other embodiments, the number of the memory cells **10** can be greater than three and arranged as an array.

(49) In FIGS. **21**B to **21**C, the isolation features **146** and **226** in the transition region **116** are connected and form an isolation feature. The substrate **110** has a protrusion portion **116***p* between a first portion and a second portion of the isolation feature (e.g., between the isolation features **146** and **226**). In some embodiments, a top surface **146**T of the first portion of the isolation feature (e.g., the isolation feature **146**) has a first part **146**TA and a second part **146**TB between the first part **146**TA and the second portion of the isolation feature (e.g., the isolation feature **226**). The second part **146**TB may be lower than the first part **146**TA. For example, the second part **146**TB may be substantially level with a top surface **226**T of the second portion of the isolation feature (e.g., the isolation feature **226**) and a top surface **222**T of the isolation feature **222**. In some embodiments, the first part **146**TA may be substantially level with a top surface **144**T of the isolation feature **144**.

(50) In some embodiments, a bottom surface **146**B of the first portion of the isolation feature (e.g., the isolation feature **146**) and a bottom surface **226**B of the second portion of the isolation feature (e.g., the isolation feature **226**) are at different levels. In some embodiments, the dummy semiconductor device **400**′ is over the first part **146**TA of the top surface **146**T of the first portion of the isolation feature (e.g., the isolation feature **146**), and the semiconductor stack **300**″ is over the second portion of the isolation feature (e.g., the isolation feature **226**).

(51) FIGS. **22**A to **22**B are cross-sectional views of a semiconductor device in accordance with some embodiments. The cross-sectional positions of FIGS. **22**A and **22**B are respectively the same as the cross-sectional position of FIGS. **21**B and **21**C. The difference between the semiconductor device of FIGS. **22**A to **22**B and the semiconductor device of FIGS. **21**B to **21**C pertains to the profile of the isolation features **146** and **226**. In the present embodiments, the trench **116**T′ (as shown in FIGS. **10**A to **10**C) expose the entire sidewalls of the isolation feature **146**, and the isolation feature **226** formed in the trench **116**T′ is in contact with the sidewall of the isolation feature **146** without any portion of the substrate **110** intervening therebetween. The bottom surface **226**B of the isolation feature **226** is higher than the bottom surface **146**B of the isolation feature **146**, and a step S is formed between the bottom surfaces **226**B and **146**B. Other relevant structural details of the semiconductor device of FIGS. **22**A to **22**B are similar to the semiconductor device of FIGS. **21**A to **21**C, and, therefore, a description in this regard will not be repeated hereinafter.

(52) FIGS. **23**A to **23**B are cross-sectional views of a semiconductor device in accordance with some embodiments. The cross-sectional positions of FIGS. **23**A and **23**B are respectively the same as the cross-sectional position of FIGS. **21**B and **21**C. The difference between the semiconductor device of FIGS. **23**A to **23**B and the semiconductor device of FIGS. **21**B to **21**C is that: in the present embodiments, the bottom surfaces **144**B and **146**B of the isolation features **144** and **146** are higher than the bottom surfaces **222**B and **226**B of the isolation features **222** and **226**. Other relevant structural details of the semiconductor device of FIGS. **23**A to **23**B are similar to the semiconductor device of FIGS. **21**A to **21**C, and, therefore, a description in this regard will not be repeated hereinafter.

(53) FIGS. **24**A to **24**B are cross-sectional views of a semiconductor device in accordance with some embodiments. The cross-sectional positions of FIGS. **24**A and **24**B are respectively the same as the cross-sectional position of FIGS. **21**B and **21**C. The difference between the semiconductor device of FIGS. **24**A to **24**B and the semiconductor device of FIGS. **21**B to **21**C is that: in the present embodiments, the bottom surface **146**B of the isolation feature **146** is substantially level with the bottom surfaces **222**B and **226**B of the isolation features **222** and **226**. Other relevant structural details of the semiconductor device of FIGS. **24**A to **24**B are similar to the semiconductor device of FIGS. **21**A to **21**C, and, therefore, a description in this regard will not be repeated hereinafter.

(54) The present disclosure is applicable to fabrication of an embedded flash memory. Based on the

above discussions, it can be seen that the present disclosure offers advantages. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. One advantage is that the floating gates are formed without being planarized, such that the floating gates are prevented from dishing and erosion issue caused by the planarization process, which in turn will prevent floating gates in array center from being over-polished, and thereby improving the thickness uniformity of the floating gates in the array center and array edge and increasing yield rate. Another advantage is that a patterned hard mask used for protecting the peripheral region during the substrate recessing process may also be used for protects the peripheral region in other processes (e.g., oxidizing the surface layer of the substrate to form tunneling layer, patterning the floating gate layer, forming the trenches in the cell region, and/or recessing the isolation features), thereby saving the number of masks.

(55) According to some embodiments, an integrated circuit device includes a substrate, an isolation feature, a memory cell, and a semiconductor device. The substrate has a cell region, a peripheral region, and a transition region between the cell region and the peripheral region. The isolation feature is in the transition region. A top surface of the isolation feature has a first portion and a second portion lower than the first portion, the second portion of the top surface of the isolation feature is between the cell region and the first portion of the top surface of the isolation feature, and a bottom surface of the isolation feature has a step height directly below the second portion of the top surface of the isolation feature. The is memory cell over the cell region of the substrate. The semiconductor device is over the peripheral region of the substrate.

(56) According to some embodiments, an integrated circuit device includes a substrate, an isolation feature, a memory cell, a semiconductor device, and an interlayer dielectric layer. The substrate has a cell region, a peripheral region, and a transition region between the cell region and the peripheral region. The isolation feature is in the transition region, wherein a bottom surface of the isolation feature has a step height. The memory cell is over the cell region of the substrate. The semiconductor device is over the peripheral region of the substrate. The interlayer dielectric layer is over the isolation feature. A bottom surface of the interlayer dielectric layer has a first portion and a second portion lower than the first portion of the bottom surface of the interlayer dielectric layer, and the step height of the bottom surface of the isolation feature is directly below the second portion of the bottom surface of the interlayer dielectric layer.

(57) According to some embodiments, an integrated circuit device includes a substrate having a cell region, a peripheral region, and a transition region between the cell region and the peripheral region; a first isolation feature in the transition region, wherein a bottom surface of the first isolation feature has a first portion, a second portion between the cell region and the first portion, and a step height connecting the first portion to the second portion, a top surface of the first isolation feature has a first portion, a second portion lower than the first portion of the top surface of the first isolation feature, and a connecting portion connecting the first portion of the top surface of the first isolation feature to the second portion of the top surface of the first isolation feature, and the connecting portion of the top surface of the first isolation feature vertically overlaps the first portion of the bottom surface of the first isolation feature; a memory cell over the cell region of the substrate; and a semiconductor device over the peripheral region of the substrate.

(58) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit device, comprising: a substrate having a cell region, a peripheral region, and a transition region between the cell region and the peripheral region; an isolation feature in the transition region, wherein a top surface of the isolation feature has a first portion and a second portion lower than the first portion, the second portion of the top surface of the isolation feature is between the cell region and the first portion of the top surface of the isolation feature, and a bottom surface of the isolation feature has a step directly below the second portion of the top surface of the isolation feature; a memory cell over the cell region of the substrate; and a semiconductor device over the peripheral region of the substrate.
2. The integrated circuit device of claim 1, wherein the bottom surface of the isolation feature has a first portion and a second portion higher than the first portion of the bottom surface of the isolation feature, and the second portion of the bottom surface of the isolation feature is between the cell region and the first portion of the bottom surface of the isolation feature.
3. The integrated circuit device of claim 1, wherein a top surface of the cell region of the substrate is lower than a top surface of the peripheral region of the substrate.
4. The integrated circuit device of claim 1, wherein the second portion of the top surface of the isolation feature is lower than a top surface of the peripheral region of the substrate and higher than a top surface of the cell region of the substrate.
5. The integrated circuit device of claim 1, further comprising: a dummy semiconductor device over the first portion of the top surface of the isolation feature; a semiconductor stack over the second portion of the top surface of the isolation feature; and an interlayer dielectric layer over the top surface of the isolation feature, wherein the interlayer dielectric layer has a portion between the dummy semiconductor device and the semiconductor stack, and the step of the bottom surface of the isolation feature is directly below the portion of the interlayer dielectric layer.
6. The integrated circuit device of claim 1, wherein the top surface of the isolation feature has a connecting portion connecting the first portion of the top surface of the isolation feature to the second portion of the top surface of the isolation feature, and the connecting portion of the top surface of the isolation feature vertically overlaps a portion of the bottom surface of the isolation feature between the step and the peripheral region.
7. The integrated circuit device of claim 1, further comprising: an etch stop layer extending over the first and second portions of the top surface of the isolation feature.
8. The integrated circuit device of claim 1, wherein the isolation feature has an interface extending from the step of the bottom surface of the isolation feature to the second portion of the top surface of the isolation feature.
9. An integrated circuit device, comprising: a substrate having a cell region, a peripheral region, and a transition region between the cell region and the peripheral region; an isolation feature in the transition region, wherein a bottom surface of the isolation feature has a step; a memory cell over the cell region of the substrate; a semiconductor device over the peripheral region of the substrate; and an interlayer dielectric layer over the isolation feature, wherein a bottom surface of the interlayer dielectric layer has a first portion and a second portion lower than the first portion of the bottom surface of the interlayer dielectric layer, and the step of the bottom surface of the isolation feature is directly below the second portion of the bottom surface of the interlayer dielectric layer.
10. The integrated circuit device of claim 9, further comprising: a dummy semiconductor device over the isolation feature; and a semiconductor stack over the isolation feature, wherein the interlayer dielectric layer spaces the semiconductor stack from the dummy semiconductor device.
11. The integrated circuit device of claim 10, wherein the semiconductor stack comprises a same gate electrode material as a select gate of the memory cell.
12. The integrated circuit device of claim 10, wherein the dummy semiconductor device comprises

a same metal material as the semiconductor device.
13. The integrated circuit device of claim 9, further comprising: an etch stop layer spacing the first and second portions of the bottom surface of the interlayer dielectric layer from the isolation feature.
14. The integrated circuit device of claim 13, wherein the isolation feature has an interface extending from the step of the bottom surface of the isolation feature to the etch stop layer.
15. An integrated circuit device, comprising: a substrate having a cell region, a peripheral region, and a transition region between the cell region and the peripheral region; a first isolation feature in the transition region, wherein a bottom surface of the first isolation feature has a first portion, a second portion between the cell region and the first portion, and a step connecting the first portion to the second portion, a top surface of the first isolation feature has a first portion, a second portion lower than the first portion of the top surface of the first isolation feature, and a connecting portion connecting the first portion of the top surface of the first isolation feature to the second portion of the top surface of the first isolation feature, and the connecting portion of the top surface of the first isolation feature vertically overlaps the first portion of the bottom surface of the first isolation feature; a memory cell over the cell region of the substrate; and a semiconductor device over the peripheral region of the substrate.
16. The integrated circuit device of claim 15, wherein the first and second portions of the bottom surface of the first isolation feature are at different levels.
17. The integrated circuit device of claim 15, further comprising: a dummy semiconductor device over the first portion of the top surface of the first isolation feature, wherein the dummy semiconductor device comprises a same metal material as the semiconductor device.
18. The integrated circuit device of claim 15, further comprising: a semiconductor stack over the second portion of the top surface of the first isolation feature, wherein the semiconductor stack comprises a same gate electrode material as a select gate of the memory cell.
19. The integrated circuit device of claim 15, further comprises: a second isolation feature in the cell region, wherein a bottom surface of the second isolation feature is substantially level with the second portion of the bottom surface of the first isolation feature.
20. The integrated circuit device of claim 15, further comprises: a third isolation feature in the peripheral region, wherein a bottom surface of the third isolation feature is substantially level with the first portion of the bottom surface of the first isolation feature.